

ACME-YOLO: An Enhanced Small Object Detection Framework in UAVs

Architecture	Performance/mAP50						
	VisDrone2019		UAVDT		TinyPerson		DOTAv1
	Val	Test	Val	Test	Val	Test	Val
YOLOv10-N	0.328	0.261	0.845	0.876	0.225	0.197	0.422
ACME-YOLO	0.347 (+0.019)	0.285 (+0.024)	0.858 (+0.013)	0.890 (+0.014)	0.243 (+0.018)	0.212 (+0.015)	0.441 (+0.019)

Architecture	Efficiency				
	Input	Params/M	GFLOPs	FPS	Size/MB
YOLOv10-N	640	2.7	8.4	187	5.9
ACME-YOLO	640	3.0	8.7	182	6.3

Table 1. Comparison of YOLOv10-N and ACME-YOLO on VisDrone2019, UAVDT, TinyPerson, and DOTAv1 with efficiency. Progressively deeper colors highlight improvements ($\Delta \geq 0.005$ mAP_{50}).

► To All Reviewers / Area Chair:

We sincerely thank all reviewers for their constructive feedback and have carefully revised the manuscript accordingly.

Q1: Concern regarding Incremental Novelty / Engineering Combination

A: We respectfully clarify that ACME-YOLO is not a mere combination of existing techniques. While basic operators like attention are established, our core contribution is a **constraint-aware structural co-design** tailored for the extreme accuracy-efficiency trade-off in UAV detection. Instead of naively stacking heavy modules, we resolve mutually coupled bottlenecks via a tightly optimized **enhancement-fusion-restoration** pipeline: Specifically, **CCRE** embeds iAFF into a dual-path residual structure with a learnable scaling factor α , enhancing shallow-layer discriminability for minute objects while strictly bounding parameter bloat. Simultaneously, **ADRF-Fusion** abandons expensive multi-scale modules in favor of parallel depth-wise dilated convolutions with Softmax weighting, achieving adaptive receptive field fusion with near-zero overhead. As proven in ablation studies, rather than being isolated add-ons, these components form a highly synergistic closed-loop. Therefore, our novelty is not claimed at the basic operator level, but rather in establishing a **lightweight architectural paradigm** that successfully bridges the gap between high-precision small-object feature recovery and stringent edge-deployment efficiency.

► Reviewer#6.

Q1: Concern on Limited Performance Gain and Robustness

A: To demonstrate robustness and generalization, we have expanded our evaluation to two additional challenging datasets (TinyPerson and DOTAv1) in Tab. 1, where ACME-YOLO consistently outperforms the baseline.

► Reviewer#7. (Addressed in the earlier responses.)

Architecture	Loc	Miss	Both	FalsePos	FalseNeg	APs
YOLOv10-N	3.54	6.44	0.25	13.21	26.46	7.69
ACME-YOLO	3.30(↓)	6.15(↓)	0.23(↓)	13.08(↓)	24.37(↓)	8.84 (+1.15)

Table 2. TIDE error decomposition with small objects mAP. Values are reported for YOLOv10-N and ACME-YOLO.

► Reviewer#8.

Q1: Concern on the rationality of "Adaptive" in ADRF

A: We respectfully clarify that "Adaptive" denotes **data-driven, task-level adaptation** rather than costly instance-wise gating. During training, the Softmax-normalized weights automatically adapt to the UAV dataset's scale distribution, prioritizing optimal receptive fields for small objects without heavy computational overhead of dynamic attention.

Q2: Concern on missing FPS/latency and deployment results

A: The native PyTorch inference speed on an A100 GPU is reported in Tab. 1 (182 FPS). We are currently finalizing the TensorRT/ONNX deployment pipelines and will update the comprehensive edge-hardware latency in the final version.

Q3: Concern on insufficient small-object evaluation metrics

A: As presented in Tab. 2, we have provided APs (small-object AP) to evaluate small object detection performance, together with TIDE analysis.

Q4: Concern on inconsistent training settings among models

A: All compared variants are trained with the same training strategy, input size, and augmentations till convergence.

► Reviewer#9.

Q1: Concern on selective metric reporting (mAP_{50} vs. $mAP_{50:95}$)

A: We clarify that mAP_{50} is the standard metric for ultra-dense datasets like VisDrone. High IoU thresholds overly penalize inherent bounding-box ambiguities via matching noise rather than true misdetections. Regardless, ACME-YOLO still improves $mAP_{50:95}$, and the newly added AP_S (Tab. 2) explicitly confirms robust gains on tiny targets.

Q2: Concern on missing deep analysis and failure modes

A: TIDE analysis (Tab. 2) mathematically details *why* our modules work: (1) **Loc Error** drops, proving Adaptive-UpSample preserves spatial details. (2) **Miss & FalseNeg Errors** decrease, showing CCRE and LDRF-Fusion effectively extract weak features. Consequently, AP_S improves by +1.15, explicitly resolving the targeted UAV bottlenecks.

► Reviewer#10.

Q1: Typos, small figure fonts, and formatting issues

A: Thanks for careful check. We have carefully corrected typos, enlarged fonts in figures, and polished formatting.